# OpenEdge Bridge — Documentation

## Sample INSERT Data

### MySQL — `id_print` (11 records)

```
INSERT INTO id_print (batch_no, batch_date, app_no, dob, cp_no, fullname, hcp_fullname, address, hcp_cpno, lname, fname, mname, spa_docno, page_no, book_no, series_no, spa_atty, spa_dt_issue, printed, printed_dt, printed_tm, tran_type, id_code, sfx, photo, gender, date_issued, id_no, year_applied, control_no, print_count) VALUES (1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr. Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 'no', NULL, NULL, 1, 5001, '', NULL, 1, '2026-01-15', 10001, 2026, 1, 0), (1, '2026-01-15', 1002, '1985-08-12', 'CP-002', 'Maria Clara Reyes', 'Dr. Jose Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes', 'Maria', 'Clara', 'SPA-002', 'P002', 'BK001', 'SR002', 'Atty. Cruz', '2025-12-02', 'no', NULL, NULL, 1, 5002, '', NULL, 2, '2026-01-15', 10002, 2026, 2, 0), (1, '2026-01-16', 1003, '1992-03-25', 'CP-003', 'Pedro Penduko', 'Dr. Ana Lopez', '789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko', 'Pedro', 'Garcia', 'SPA-003', 'P003', 'BK002', 'SR003', 'Atty. Santos', '2025-12-03', 'no', NULL, NULL, 2, 5003, 'JR', NULL, 1, '2026-01-16', 10003, 2026, 3, 1), (2, '2026-02-10', 1004, '1988-11-30', 'CP-004', 'Ana Patricia Gomez', 'Dr. Luis Torres', '321 Aguinaldo Rd, Davao', 'HCP-004', 'Gomez', 'Ana', 'Patricia', 'SPA-004', 'P004', 'BK002', 'SR004', 'Atty. Reyes', '2025-12-10', 'yes', '2026-02-11', '10:30:00', 1, 5004, '', NULL, 2, '2026-02-10', 10004, 2026, 4, 1), (2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel Tan', 'Dr. Elena Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan', 'Carlos', 'Miguel', 'SPA-005', 'P005', 'BK003', 'SR005', 'Atty. Dela Rosa', '2025-12-15', 'no', NULL, NULL, 3, 5005, '', NULL, 1, '2026-02-10', 10005, 2026, 5, 2), (2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia Isabelle Lim', 'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim', 'Sofia', 'Isabelle', 'SPA-006', 'P006', 'BK003', 'SR006', 'Atty. Mercado', '2025-12-20', 'no', NULL, NULL, 1, 5006, '', NULL, 2, '2026-02-11', 10006, 2026, 6, 0), (3, '2026-03-05', 1007, '1998-09-18', 'CP-007', 'Rafael Antonio Uy', 'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy', 'Rafael', 'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty. Garcia', '2026-01-05', 'yes', '2026-03-06', '14:15:00', 2, 5007, '', NULL, 1, '2026-03-05', 10007, 2026, 7, 1), (3, '2026-03-05', 1008, '1991-12-03', 'CP-008', 'Gabriela Nicole Sy', 'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig', 'HCP-008', 'Sy', 'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty. Ramos', '2026-01-10', 'no', NULL, NULL, 1, 5008, '', NULL, 2, '2026-03-05', 10008, 2026, 8, 0), (3, '2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co', 'Dr. Teresa Javier', '246 Osmena St, Baguio', 'HCP-009', 'Co', 'Miguel', 'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty. Villar', '2026-01-15', 'no', NULL, NULL, 3, 5009, 'III', NULL, 1, '2026-03-06', 10009, 2026, 9, 3), (4, '2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua', 'Dr. Roberto Domingo', '135 Kalayaan Ave, Pampanga', 'HCP-010', 'Chua', 'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005', 'SR010', 'Atty. Pineda', '2026-02-01', 'no', NULL, NULL, 1, 5010, '', NULL, 2, '2026-04-20', 10010, 2026, 10, 0), (4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis Pangilinan', 'Dr. Carmen Salazar', '864 EDSA, Mandaluyong', 'HCP-011', 'Pangilinan', 'Antonio', 'Luis', 'SPA-011', 'P011', 'BK006', 'SR011', 'Atty. Hernandez', '2026-02-10', 'yes', '2026-04-21', '09:00:00', 2, 5011, '', NULL, 1, '2026-04-20', 10011, 2026, 11, 2);
```

### OpenEdge (Progress) — `pub."id-print"` (11 records)

```
INSERT INTO pub."id-print" ("batch-no", "batch-date", "app-no", "dob", "cp-no", "fullname", "hcp-fullname", "address", "hcp-cpno", "lname", "fname", "mname", "spa-docno", "page-no", "book-no", "series-no", "spa-atty", "spa-dt-issue", "printed", "printed-dt", "printed-tm",
```

```
"tran-type", "id-code", "sfx", "photo", "gender", "date-issued", "id-no", "year-applied",
"control-no") VALUES (1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr. Maria
Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos', 'SPA-001', 'P001',
'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 0, NULL, NULL, 1, 5001, '', NULL, 1, '2026-01-15',
10001, 2026, 1), (1, '2026-01-15', 1002, '1985-08-12', 'CP-002', 'Maria Clara Reyes', 'Dr. Jose
Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes', 'Maria', 'Clara', 'SPA-002', 'P002',
'BK001', 'SR002', 'Atty. Cruz', '2025-12-02', 0, NULL, NULL, 1, 5002, '', NULL, 2, '2026-01-15',
10002, 2026, 2), (1, '2026-01-16', 1003, '1992-03-25', 'CP-003', 'Pedro Penduko', 'Dr. Ana Lopez',
'789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko', 'Pedro', 'Garcia', 'SPA-003', 'P003', 'BK002',
'SR003', 'Atty. Santos', '2025-12-03', 0, NULL, NULL, 2, 5003, 'JR', NULL, 1, '2026-01-16', 10003,
2026, 3), (2, '2026-02-10', 1004, '1988-11-30', 'CP-004', 'Ana Patricia Gomez', 'Dr. Luis Torres',
'321 Aguinaldo Rd, Davao', 'HCP-004', 'Gomez', 'Ana', 'Patricia', 'SPA-004', 'P004', 'BK002',
'SR004', 'Atty. Reyes', '2025-12-10', 1, '2026-02-11', '10:30:00', 1, 5004, '', NULL, 2,
'2026-02-10', 10004, 2026, 4), (2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel
Tan', 'Dr. Elena Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan', 'Carlos', 'Miguel',
'SPA-005', 'P005', 'BK003', 'SR005', 'Atty. Dela Rosa', '2025-12-15', 0, NULL, NULL, 3, 5005, '',
NULL, 1, '2026-02-10', 10005, 2026, 5), (2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia
Isabelle Lim', 'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim', 'Sofia',
'Isabelle', 'SPA-006', 'P006', 'BK003', 'SR006', 'Atty. Mercado', '2025-12-20', 0, NULL, NULL, 1,
5006, '', NULL, 2, '2026-02-11', 10006, 2026, 6), (3, '2026-03-05', 1007, '1998-09-18', 'CP-007',
'Rafael Antonio Uy', 'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy', 'Rafael',
'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty. Garcia', '2026-01-05', 1, '2026-03-06',
'14:15:00', 2, 5007, '', NULL, 1, '2026-03-05', 10007, 2026, 7), (3, '2026-03-05', 1008,
'1991-12-03', 'CP-008', 'Gabriela Nicole Sy', 'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig',
'HCP-008', 'Sy', 'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty. Ramos',
'2026-01-10', 0, NULL, NULL, 1, 5008, '', NULL, 2, '2026-03-05', 10008, 2026, 8), (3,
'2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co', 'Dr. Teresa Javier', '246 Osmena
St, Baguio', 'HCP-009', 'Co', 'Miguel', 'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty.
Villar', '2026-01-15', 0, NULL, NULL, 3, 5009, 'III', NULL, 1, '2026-03-06', 10009, 2026, 9), (4,
'2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua', 'Dr. Roberto Domingo', '135
Kalayaan Ave, Pampanga', 'HCP-010', 'Chua', 'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005',
'SR010', 'Atty. Pineda', '2026-02-01', 0, NULL, NULL, 1, 5010, '', NULL, 2, '2026-04-20', 10010,
2026, 10), (4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis Pangilinan', 'Dr. Carmen
Salazar', '864 EDSA, Mandaluyong', 'HCP-011', 'Pangilinan', 'Antonio', 'Luis', 'SPA-011', 'P011',
'BK006', 'SR011', 'Atty. Hernandez', '2026-02-10', 1, '2026-04-21', '09:00:00', 2, 5011, '', NULL,
1, '2026-04-20', 10011, 2026, 11);
```

**Version:** 1.2
**Date:** September 4, 2026
**Purpose:** Sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing. Reads `"id-print"` from OpenEdge and mirrors it to MySQL's `id_print` table.

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## Architecture Overview

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Progress OpenEdge SQL Server ■ ← Source of truth
(remote, provided by client) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ODBC
■■■■■■■■■■■■■■■■■■■
```

```
■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■ ■ Entrust Instant ID ■ ← Card issuance system
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

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## Data Flow

### OpenEdge → MySQL (One-Way Sync)

```
OpenEdge "id-print" ■■■ C# Service ■■■ MySQL id_print table (read) (sync loop) (upsert)
```

All 31 fields from the `id-print` table are synced, including:
- `batch-no`, `batch-date`, `app-no`, `dob`
- `cp-no`, `fullname`, `hcp-fullname`, `address`, `hcp-cpno`
- `lname`, `fname`, `mname`
- `spa-docno`, `page-no`, `book-no`, `series-no`, `spa-atty`, `spa-dt-issue`
- `printed`, `printed-dt`, `printed-tm`
- `tran-type`, `image-data`, `id-code`, `sfx`, `photo`
- `gender`, `date-issued`, `id-no`, `year-applied`, `control-no`

Runs every N seconds (configurable in `appsettings.json`). Uses **check-then-update** upserts (SELECT by `app_no`, then UPDATE or INSERT) to avoid duplicates. `print_count` is only inserted on first sync; subsequent updates do NOT overwrite it (Instant ID manages `print_count`).

#### Full vs. Incremental Sync

The bridge has two read modes, decided automatically on every cycle:

| Mode | Trigger | What it reads from OpenEdge |
|---|---|---|
| **FULL** | No `last_full_sync_utc` recorded yet (first run, or fresh MySQL) | All rows in `pub."id-print"` — seeds/refreshes the entire MySQL table |
| **INCREMENTAL** | A full sync has already been recorded | Only rows with `batch-no` greater than the highest `batch_no` already in MySQL |

OpenEdge `id-print` records are **write-once** — new data arrives only as new batches; existing rows are never edited. Therefore, with the default `Sync:FullSyncIntervalMinutes = 0`, the full sync runs **once** (the initial seed) and every later cycle is incremental. Restarting the service or rebooting the PC does **not** cause another full sync — the timestamp is persisted in MySQL's `sync_state` table (see Database Tables).

If `Sync:FullSyncIntervalMinutes` is set to a positive value (e.g., `60`), a full sync also runs when that many minutes have elapsed since the last successful full sync — useful only if OpenEdge rows ever start getting edited in place.

> **Note:** `signature_data` and `qr_data` are **not** part of this sync — they are MySQL-only columns populated by Entrust Instant ID and never touched by the bridge (see Database Tables).

### MySQL → OpenEdge (Printed Back-Sync)

```
MySQL id_print ■■■■■■■■ C# Service ■■■■■■■■ OpenEdge "id-print" (print_count > 0) (check + mark) (SET printed = yes)
```

When Instant ID increments `print_count > 0` on a record, the bridge detects it and sets `printed = yes` in the corresponding OpenEdge record.

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## Prerequisites

### On the Target Windows PC

| # | Component | Version / File | Notes |
|---|---|---|---|
| 1 | **Windows OS** | Windows 10/11 or Windows Server 2016+ | 64-bit required |
| 2 | **OpenEdge ODBC Driver** | `PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe` | Install, configure a System DSN |
| 3 | **XAMPP** | Latest (8.x) | Provides local MySQL server |
| 4 | **MySQL Connector/J** | `mysql-connector-j-8.0.33.jar` | For Entrust Instant ID to connect to MySQL |
| 5 | **OpenEdgeBridge** | Published `publish/` folder | Self-contained, no .NET runtime needed |

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## Local Development — Installing Progress OpenEdge Database

> **Use this section when no remote OpenEdge server is available.**
> You can install Progress OpenEdge locally, create a database, and run the SQL below.

### Step 0-A: Download & Install Progress OpenEdge

1. Go to https://www.progress.com/openedge/downloads
2. Download **Progress OpenEdge Developer Studio** or **Progress OpenEdge Personal Edition** (free for development)
3. Run the installer (typically `PROGRESS_OE_12.x_WIN_64.exe`) as Administrator
4. Choose **Typical** or **Complete** installation
5. Default install path: `C:\Progress\OpenEdge\`

After installation, a **"Proenv"** shortcut will appear in your Start Menu.
**Proenv** is a command prompt pre-configured with all OpenEdge environment variables.

## Step 0-B: Create a Local OpenEdge Database

Open **Proenv** as Administrator and run:

```
:: Create a folder for your database mkdir C:\OpenEdgeDatabases cd C:\OpenEdgeDatabases :: Create an empty database called 'iddb' prodb iddb empty
```

This creates `iddb.db` and related files in `C:\OpenEdgeDatabases\`.

## Step 0-C: Start the Database Server

In **Proenv**, run:

```
:: Start the database for multi-user access with both 4GL and SQL enabled proserve "C:\OpenEdgeDatabases\iddb.db" -S 11918 -ServerType Both -n 51 -L 8192
```

| Parameter | Meaning |
|---|---|
| `-S 11918` | Service port (SQL and 4GL clients connect on this port) |
| `-n 51` | Maximum concurrent users |
| `-L 8192` | Lock table size (entries) |
| `-ServerType Both` | **Required** — enables both 4GL and SQL-92 engine |

The database is now running. Keep this Proenv window open.

## Step 0-D: Configure the ODBC System DSN

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab, click **Add...**
3. Select **Progress OpenEdge 12.x Driver** (or "DataDirect OpenEdge SQL Server")
4. Configure:
5. **Data Source Name:** `OpenEdgeInstantID`
6. **Description:** `OpenEdge Instant ID Database`
7. **Host Name:** `localhost`
8. **Port Number:** `11918`
9. **Database Name:** `iddb`
10. **User ID:** `admin`
11. Click **Test Connect** → should succeed
12. Click **OK**

## Step 0-E: Create the Database Schema (`.df` File)

The table schema is defined in a `.df` (Data Definition) file. Create a file named `iddb.df` in `C:\OpenEdgeDatabases\` with the following content:

```
ADD TABLE "id-print" AREA "Schema Area" DUMP-NAME "id-print" ADD FIELD "batch-no" OF "id-print" AS
integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 2 MAX-WIDTH 4 ORDER 10 ADD FIELD "batch-date" OF
"id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 3 MAX-WIDTH 4 ORDER 20 ADD FIELD
"app-no" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 4 MAX-WIDTH 4 ORDER 30
ADD FIELD "dob" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 5 MAX-WIDTH 4 ORDER
40 ADD FIELD "cp-no" OF "id-print" AS character FORMAT "x(15)" INITIAL "" POSITION 6 MAX-WIDTH 30
ORDER 50 ADD FIELD "fullname" OF "id-print" AS character FORMAT "x(65)" INITIAL "" POSITION 7
MAX-WIDTH 130 ORDER 60 ADD FIELD "hcp-fullname" OF "id-print" AS character FORMAT "x(65)" INITIAL
"" POSITION 8 MAX-WIDTH 130 ORDER 70 ADD FIELD "address" OF "id-print" AS character FORMAT "x(65)"
INITIAL "" POSITION 9 MAX-WIDTH 130 ORDER 80 ADD FIELD "hcp-cpno" OF "id-print" AS character
FORMAT "x(15)" INITIAL "" POSITION 10 MAX-WIDTH 30 ORDER 90 ADD FIELD "lname" OF "id-print" AS
character FORMAT "x(55)" INITIAL "" POSITION 11 MAX-WIDTH 110 ORDER 100 ADD FIELD "fname" OF
"id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 12 MAX-WIDTH 110 ORDER 110 ADD FIELD
"mname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 13 MAX-WIDTH 110 ORDER 120
ADD FIELD "spa-docno" OF "id-print" AS character FORMAT "x(25)" INITIAL "" POSITION 14 MAX-WIDTH
50 ORDER 130 ADD FIELD "page-no" OF "id-print" AS character FORMAT "x(20)" INITIAL "" POSITION 15
MAX-WIDTH 40 ORDER 140 ADD FIELD "book-no" OF "id-print" AS character FORMAT "x(20)" INITIAL ""
POSITION 16 MAX-WIDTH 40 ORDER 150 ADD FIELD "series-no" OF "id-print" AS character FORMAT "x(20)"
INITIAL "" POSITION 17 MAX-WIDTH 40 ORDER 160 ADD FIELD "spa-atty" OF "id-print" AS character
FORMAT "x(65)" INITIAL "" POSITION 18 MAX-WIDTH 130 ORDER 170 ADD FIELD "spa-dt-issue" OF
"id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 19 MAX-WIDTH 4 ORDER 180 ADD FIELD
"printed" OF "id-print" AS logical FORMAT "yes/no" INITIAL "no" POSITION 20 MAX-WIDTH 1 ORDER 190
ADD FIELD "printed-dt" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 21 MAX-WIDTH 4
ORDER 200 ADD FIELD "printed-tm" OF "id-print" AS character FORMAT "x(35)" INITIAL "" POSITION 22
MAX-WIDTH 70 ORDER 210 ADD FIELD "tran-type" OF "id-print" AS integer DESCRIPTION "1 = for RTC,
2=senior" FORMAT "->,>>>,>>9" INITIAL "0" POSITION 23 MAX-WIDTH 4 ORDER 220 ADD FIELD "image-data"
OF "id-print" AS raw FORMAT "x(55)" INITIAL "" POSITION 24 MAX-WIDTH 110 ORDER 230 ADD FIELD
"id-code" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 25 MAX-WIDTH 4 ORDER
240 ADD FIELD "sfx" OF "id-print" AS character FORMAT "x(15)" INITIAL "" POSITION 26 MAX-WIDTH 30
ORDER 250 ADD FIELD "photo" OF "id-print" AS blob FORMAT "x(8)" INITIAL ? POSITION 27 LOB-AREA
"Schema Area" LOB-BYTES 104857600 LOB-SIZE 100M ORDER 260 ADD FIELD "gender" OF "id-print" AS
integer DESCRIPTION "1=Female, 2=Male" FORMAT "->,>>>,>>9" INITIAL "0" POSITION 28 MAX-WIDTH 4
ORDER 270 ADD FIELD "date-issued" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 29
MAX-WIDTH 4 ORDER 280 ADD FIELD "id-no" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0"
POSITION 30 MAX-WIDTH 4 ORDER 290 ADD FIELD "year-applied" OF "id-print" AS integer FORMAT
">>>9999" INITIAL "0" POSITION 31 MAX-WIDTH 4 ORDER 300 ADD FIELD "control-no" OF "id-print" AS
integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 32 MAX-WIDTH 4 ORDER 310 . PSC cpstream=ISO8859-1
. 0000004144
```

Then, in **Proenv**, load the `.df` file into the database:

```
cd C:\OpenEdgeDatabases prodict iddb
```

In the Data Dictionary menu, select **Admin** → **Load Data Definitions** → **.df file**, then choose `iddb.df`.

> **Note:** The database name is `iddb` and the schema resides under the `PUB` area (default).

---

## Database Tables

**`id-print`** / **`id_print`**

Converted from the provided `.df` schema.
**Paste each CREATE TABLE directly into your SQL editor and run it.**

### OpenEdge SQL — CREATE TABLE

```
CREATE TABLE "id-print" ( "batch-no" INTEGER NOT NULL DEFAULT 0, "batch-date" DATE NULL, "app-no"
INTEGER NOT NULL DEFAULT 0, "dob" DATE NULL, "cp-no" VARCHAR(15) NOT NULL DEFAULT '', "fullname"
VARCHAR(65) NOT NULL DEFAULT '', "hcp-fullname" VARCHAR(65) NOT NULL DEFAULT '', "address"
VARCHAR(65) NOT NULL DEFAULT '', "hcp-cpno" VARCHAR(15) NOT NULL DEFAULT '', "lname" VARCHAR(55)
NOT NULL DEFAULT '', "fname" VARCHAR(55) NOT NULL DEFAULT '', "mname" VARCHAR(55) NOT NULL DEFAULT
'', "spa-docno" VARCHAR(25) NOT NULL DEFAULT '', "page-no" VARCHAR(20) NOT NULL DEFAULT '',
"book-no" VARCHAR(20) NOT NULL DEFAULT '', "series-no" VARCHAR(20) NOT NULL DEFAULT '', "spa-atty"
VARCHAR(65) NOT NULL DEFAULT '', "spa-dt-issue" DATE NULL, "printed" BIT NOT NULL DEFAULT 0,
"printed-dt" DATE NULL, "printed-tm" VARCHAR(35) NOT NULL DEFAULT '', "tran-type" INTEGER NOT NULL
DEFAULT 0, "image-data" VARBINARY(110) NULL, "id-code" INTEGER NOT NULL DEFAULT 0, "sfx"
VARCHAR(15) NOT NULL DEFAULT '', "photo" BLOB NULL, "gender" INTEGER NOT NULL DEFAULT 0,
"date-issued" DATE NULL, "id-no" INTEGER NOT NULL DEFAULT 0, "year-applied" INTEGER NOT NULL
DEFAULT 0, "control-no" INTEGER NOT NULL DEFAULT 0 );
```

**Note:** OpenEdge requires double-quotes around field names containing hyphens (`-`).
The schema resides in the `PUB` area (default).
`BIT` represents the OpenEdge `LOGICAL` type (yes/no → 1/0).
`VARBINARY(110)` stores the raw image binary data (MAX-WIDTH 110 from the schema).
`BLOB` stores the cardholder photo (LOB-SIZE 100M from the schema).

### MySQL — CREATE TABLE

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, batch_no INT NULL,
batch_date DATE NULL, app_no INT NULL, dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65)
NULL, hcp_fullname VARCHAR(65) NULL, address VARCHAR(65) NULL, hcp_cpno VARCHAR(15) NULL, lname
VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, spa_docno VARCHAR(25) NULL,
page_no VARCHAR(20) NULL, book_no VARCHAR(20) NULL, series_no VARCHAR(20) NULL, spa_atty
VARCHAR(65) NULL, spa_dt_issue DATE NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches
OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL
COMMENT '1 = for RTC, 2 = senior', image_data LONGBLOB NULL, id_code INT NULL, sfx VARCHAR(15)
NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', date_issued DATE NULL,
id_no INT NULL, year_applied INT NULL, control_no INT NULL, signature_data LONGBLOB NULL COMMENT
'Signature image; written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)', qr_data
LONGBLOB NULL COMMENT 'QR code image; written by Entrust Instant ID (MySQL only, NOT synced from
OpenEdge)', print_count INT NULL COMMENT 'Incremented by Instant ID; triggers printed=yes
back-sync', PRIMARY KEY (id), UNIQUE INDEX idx_app_no (app_no) ) ENGINE=InnoDB DEFAULT
CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

**Note:** `id` is an internal auto-increment primary key. `app_no` is the unique business key — the bridge uses it for deduplication (it SELECTs by `app_no` first, then UPDATEs or INSERTs as needed).

**MySQL-only columns (`signature_data`, `qr_data`):** These store the cardholder's signature and QR-code images and are **MySQL only**. They do **not** exist in the OpenEdge `"id-print"` table and are **never read from or written back to OpenEdge**. Entrust Instant ID populates them directly in MySQL. The bridge deliberately leaves them untouched — it does not include them in its INSERT/UPDATE, so syncing from OpenEdge will never overwrite them.

> **Already have the table?** If you created `id_print` before these columns existed, add them with:
> `sql ALTER TABLE id_print CHANGE COLUMN id_type tran_type INT NULL COMMENT '1 = for RTC, 2 = senior', ADD COLUMN sfx VARCHAR(15) NULL AFTER id_code, ADD COLUMN photo LONGBLOB NULL AFTER sfx, ADD COLUMN gender INT NULL COMMENT '1 = Female, 2 = Male' AFTER photo, ADD COLUMN date_issued DATE NULL AFTER gender, ADD COLUMN id_no INT NULL AFTER date_issued, ADD COLUMN year_applied INT NULL AFTER id_no, ADD COLUMN control_no INT NULL AFTER year_applied, ADD COLUMN signature_data LONGBLOB NULL AFTER control_no, ADD COLUMN qr_data LONGBLOB NULL AFTER signature_data;`

> **Schema update (2026-09-03):** OpenEdge renamed `id-type` to `tran-type` and added `sfx`, `photo`, `gender`, `date-issued`, `id-no`, `year-applied`, and `control-no`. If your OpenEdge `"id-print"` table was created with the older `.df`, reload the updated `.df` (Step 0-E) or add the new fields with `ADD FIELD` in the Data Dictionary, then run the ALTER above on MySQL.

### `sync_state` (bridge state table — MySQL only, auto-created)

The bridge keeps its own bookkeeping in a tiny table called `sync_state`. **You do not need to create it manually** — the bridge runs `CREATE TABLE IF NOT EXISTS` on demand, so it appears in MySQL automatically after the first sync cycle.

```
CREATE TABLE IF NOT EXISTS sync_state ( state_key VARCHAR(64) NOT NULL PRIMARY KEY, state_value VARCHAR(64) NOT NULL, updated_utc DATETIME NOT NULL ) ENGINE=InnoDB;
```

| Key | Meaning |
|---|---|
| `last_full_sync_utc` | UTC timestamp of the last successful FULL sync. Used to decide whether the next cycle can be INCREMENTAL. Survives restarts, so a reboot never forces a redundant full sync. |

### Column Comparison — Plain English

Think of each row as **one person's ID-card application folder**. The columns are the details written on that folder.

| OpenEdge name | MySQL name | Plain English meaning |
|---|---|---|
| `batch-no` | `batch_no` | **Group number** — which batch/run of applications this person came in with (like a batch printing number) |
| `batch-date` | `batch_date` | **Date of that group** — when their batch was made |
| `app-no` | `app_no` | **Application number** — the unique file number of this person's ID application. This is the "fingerprint" that tells us which person is which |
| `dob` | `dob` | **Date of birth** |
| `cp-no` | `cp_no` | **The person's reference number** (their own control/card number) |

| | | |
|---|---|---|
| `fullname` | `fullname` | **Full name** — complete name all in one |
| `hcp-fullname` | `hcp_fullname` | **Doctor's name** — the healthcare provider attached to this person |
| `address` | `address` | **Home address** |
| `hcp-cpno` | `hcp_cpno` | **Doctor's number** — the healthcare provider's own reference number |
| `lname` | `lname` | **Last name** (surname) |
| `fname` | `fname` | **First name** |
| `mname` | `mname` | **Middle name** |
| `spa-docno` | `spa_docno` | **Permission letter number** — the number of the Special Power of Attorney (SPA) document |
| `spa-atty` | `spa_atty` | **Lawyer's name** — the attorney who signed that permission letter |
| `spa-dt-issue` | `spa_dt_issue` | **Date the permission letter was issued** |
| `page-no` | `page_no` | **Page number** — which page in the record book this person is on |
| `book-no` | `book_no` | **Book number** — which record book |
| `series-no` | `series_no` | **Series number** — the serial number of the ID |
| `tran-type` | `tran_type` | **Kind of ID / transaction type** — `1` = regular (RTC), `2` = senior citizen |
| `image-data` | `image_data` | **The person's photo** (picture file) |
| `id-code` | `id_code` | **ID code** — the code number printed on the card |
| `sfx` | `sfx` | **Name suffix** — e.g. Jr., Sr., III |
| `photo` | `photo` | **Photo** — the cardholder's photograph (BLOB image) |
| `gender` | `gender` | **Gender** — `1` = Female, `2` = Male |
| `date-issued` | `date_issued` | **Date the ID card was issued** |
| `id-no` | `id_no` | **ID number** — the person's ID number |
| `year-applied` | `year_applied` | **Year the application was filed** (e.g. `2026`) |

| | | |
|---|---|---|
| `control-no` | `control_no` | **Control number** — application/control tracking number |
| `printed` | `printed` | **Has it been printed yet?** — `yes` or `no` |
| `printed-dt` | `printed_dt` | **Date it was printed** |
| `printed-tm` | `printed_tm` | **Time it was printed** |
| — | `signature_data` | **Person's signature picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in |
| — | `qr_data` | **QR code picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in |
| — | `print_count` | **How many times it got printed** — *MySQL only*. Instant ID counts this; it's never overwritten by the bridge |
| — | `id` | **Row number** — MySQL's own internal counter. Not business data, ignore it |

**30-second summary:** The bridge copies the person's details (name, birthdate, address, photo, etc.) **from OpenEdge → MySQL**, like photocopying the folder. `app_no` is the **matchmaker** — it makes sure updates land on the right person. The three columns that exist **only in MySQL** (`signature_data`, `qr_data`, `print_count`) are filled in by the **Entrust machine**, and the bridge never touches them.

---

## Step-by-Step Setup (Remote Server)

> **If you already have a remote OpenEdge server**, follow these steps.
> **If no server is available**, go back to Local Development — Installing Progress OpenEdge Database first.

### 1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab
3. Click **Add...**
4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure:

6. **Data Source Name:** e.g., `OpenEdgeInstantID`
7. **Host:** (provided by your OpenEdge team, or `localhost` if local)
8. **Port:** (provided by your OpenEdge team, or `11918` if local)
9. **Database:** (provided by your OpenEdge team, or `iddb` if local)
10. **User ID / Password:** (provided by your OpenEdge team)
11. Click **Test Connect** to verify, then **OK**

## 2. Create the OpenEdge Tables

Connect to your OpenEdge database using any ODBC-compatible SQL client (DBeaver, HeidiSQL, etc.) and paste the SQL from the Database Tables section above:

- Run the **`"id-print"`** CREATE TABLE

### 2-B. Grant Table Permissions to the Login User

The bridge connects to OpenEdge using the **User ID** configured in the ODBC DSN (e.g., `admin`). If your login user is **not** `admin`, it needs explicit permissions on the table — otherwise the bridge fails with an access-denied error when it starts.

Run the following from a DBA account (e.g., `admin`), replacing `your_login_user` with the actual user you put in the DSN:

```
GRANT SELECT, INSERT, UPDATE, DELETE ON pub."id-print" TO your_login_user;
```

> **Note:** `SELECT` is required to read records for the OpenEdge → MySQL sync, and `UPDATE` is required for the printed back-sync (`printed = yes`). `INSERT`/`DELETE` are included for completeness. `pub` is the default schema area, and the double quotes around `"id-print"` are required because of the hyphen in the table name.

## 3. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings
3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

## 4. Create the MySQL Database & Tables

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE instantid_bridge;
```

Then paste and run the MySQL CREATE TABLE statements from the Database Tables section above:

- Run the **`id_print`** CREATE TABLE

## 5. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder
2. Configure Entrust to point to the local MySQL database:
3. **Host:** `localhost`
4. **Port:** `3306`
5. **Database:** `instantid_bridge`
6. **User:** `root` (or a dedicated user)
7. **Table:** `id_print`

### 6. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\`.

**Edit `appsettings.json`**

```
{ "ConnectionStrings": { "OpenEdge": "DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;", "MySql": "Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_password;" }, "Sync": { "IntervalSeconds": 10, "FullSyncIntervalMinutes": 0 } }
```

| Setting | Description |
|---|---|
| `OpenEdge` | ODBC DSN name, user, and password |
| `MySql` | MySQL server, database, user, and password |
| `Sync:IntervalSeconds` | How often the sync loop runs (default: 10s) |
| `Sync:FullSyncIntervalMinutes` | Minutes between periodic FULL syncs. `0` = disabled (recommended; OpenEdge rows are write-once, so only the first run seeds). Positive value (e.g., `60`) = re-read the whole table that often, catching in-place edits if they ever happen |

### 7. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" ` -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" ` -DisplayName "OpenEdge Bridge Service" ` -Description "Syncs id-print data between OpenEdge and MySQL for Entrust Instant ID" ` -StartupType Automatic Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge # Status should be: Running
```

---

## Service Management Commands

```
Start-Service OpenEdgeBridge # Start Stop-Service OpenEdgeBridge # Stop Restart-Service OpenEdgeBridge # Restart (after config changes) Get-Service OpenEdgeBridge # Check status # Uninstall Stop-Service OpenEdgeBridge sc delete OpenEdgeBridge
```

---

# Logging & Troubleshooting

## Log Locations

| Source | Location |
|---|---|
| File logs | `C:\OpenEdgeBridge\Logs\` (dated log files) |
| Windows Event Viewer | *Windows Logs* → *Application*, Source: `OpenEdgeBridge` |

## Common Issues

| Problem | Likely Cause | Fix |
|---|---|---|
| Service starts then stops immediately | Wrong DSN name or credentials | Test the ODBC DSN in ODBC Administrator with **Test Connect** |
| "MySQL connection string not found" | `appsettings.json` not in same folder as `.exe` | Ensure `appsettings.json` is in `C:\OpenEdgeBridge\` |
| Duplicate rows in MySQL | Missing UNIQUE index | Run the `ALTER TABLE ... ADD UNIQUE INDEX` SQL above |
| OpenEdge ODBC driver not found | Wrong bitness (32 vs 64) | Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64 |
| `Failure getting schema read lock (12138)` | SQL engine not initialized for database | 1. Verify `-ServerType Both` or `-ServerType SQL` on proserve<br>2. In sqlexp run: `UPDATE TABLE STATISTICS AND INDEX STATISTICS AND ALL COLUMN STATISTICS;`<br>3. If still failing, restart DB: `proshut` then `proserve` with `-SQLTrc 3` |
| MySQL not accessible from Entrust | MySQL not running or wrong port | Check XAMPP Control Panel; MySQL must be **Started** |
| OpenEdge SQL CREATE TABLE fails with "Invalid table name" | Hyphens in table/field names not quoted | Wrap names containing hyphens in double quotes: `"id-print"`, `"batch-no"` |
| `image-data` field too small | VARBINARY length insufficient | The OpenEdge schema specifies MAX-WIDTH 110; adjust as needed |

| | | |
|---|---|---|
| Full sync on every start (log says `[FULL]`) | `sync_state` missing or MySQL unreachable at startup | The bridge auto-creates `sync_state`; after the first successful full sync it records a timestamp and later starts use `[INCREMENTAL]`. If MySQL is down at startup it falls back to FULL — that is safe and self-corrects once MySQL is reachable |

---

## Sync Interval Tuning

### `IntervalSeconds` — how often the bridge checks for new batches

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": { "IntervalSeconds": 30 }
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

The interval is an **idle pause between cycles** (work → wait → work), not a fixed wall-clock schedule. A slow sync simply delays the next check; cycles never overlap.

### `FullSyncIntervalMinutes` — how often the whole table is re-read

```
"Sync": { "FullSyncIntervalMinutes": 0 }
```

| Value | Behavior |
|---|---|
| `0` (default, recommended) | FULL sync only on the **first run** (seeds MySQL). All later cycles are INCREMENTAL (`batch-no` > highest in MySQL). Restarts never re-trigger a full sync. |
| Positive (e.g., `60`) | Additionally re-reads the entire table when that many minutes have passed since the last full sync. Only needed if OpenEdge rows can be edited in place. |

**How the decision works:** after every successful FULL sync the bridge writes the completion time to `sync_state.last_full_sync_utc` in MySQL. On each cycle it compares that timestamp against `FullSyncIntervalMinutes`. If MySQL is unreachable at startup, the bridge falls back to FULL — the safe default that guarantees a complete copy.

**Why `0` is right for this deployment:** OpenEdge `id-print` data is batch-based and write-once — new records always arrive in new batches (higher `batch-no`), and existing rows are not edited. The incremental query therefore sees every new record, and a periodic full re-read only wastes database load on a table that never changes. If the provider ever starts correcting existing rows, set this to `60` (hourly refresh) or similar.

---

## Port Summary

| Component | Port | Direction |
|---|---|---|
| OpenEdge SQL Server | Provided by OpenEdge team | Outbound |
| MySQL (XAMPP) | 3306 | Local |
| Entrust → MySQL | 3306 | Local via JDBC |

No inbound ports are opened by the bridge itself.

---

## Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

> **After upgrading from v1.1 or earlier:** the new `sync_state` table does not exist yet, so the first run performs one FULL sync to seed MySQL and record the timestamp. From then on, only new batches are synced — including across restarts.

---

## Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)
- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database created in MySQL
- [ ] `id-print` / `id_print` table created in **both** OpenEdge and MySQL
- [ ] `sync_state` table present in MySQL (auto-created by the bridge after the first run)
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors